

• Section A

34/50

1) What is a string in python?

→ String is a sequence of character and we used single (') and double (") quotes.

2) Which symbol is used to create a tuple?

→ Parentheses ()

3) Which python data type stores unique values?

→ Set()

4) Write the syntax to create an empty dictionary.

→ Syntax →

dict = {}

5) Which loop is generally used to iterate over a sequence?

→ for loop.

6) Which keyword is used to exit a loop immediately?

→ break keyword

7) Can a list contain duplicate values?

→ yes..

8) Can a tuple be modified after it is created?

→ No, because it is immutable.

9) What is the index of the first ele of a python list

→ 0.

10) Which function is used to find the no. of ele's in a string, list, tuple, set, or dictionary.

→ len()

• Section B - 2 Marks que.

1). Differentiate betⁿ list and tuple.

→: list tuple

1) mutable Immutable

2) [] ()

3) it can store duplicate values it can also store duplicate values.

4) list start from index 0. Tuple it start from index 0.

②

2). Diff. betⁿ set and dictionary.

→: set Dictionary

1) mutable mutable

2) { } { }

3) set stores unique values Dictionary stores unique key.

4) set is a collectⁿ of unique elements Dictionary is collectⁿ of key-value pairs

⑤

3) Write a python program to count the number of characters in a string.

→: string = "python"
 print(len(string))

using for loop

①

4). Write a python program using a for loop to print numbers from 1 to 10.

→ for i in range(1,11):
 print(i)

5) Write a python program using a while loop to print numbers from 10 to 1.

→ i = 10

while i >= 1:

 print(i)

 i = i - 1

i -= 1

• Section C - 5 M. Questions -

1) String -

Consider: text = "Python Programming",

Write a python statements to:

a) print the complete string.

→ text = "Python Programming"
 print(text)

b) print the first character.

→ text = "Python Programming"
 print(text[:5])

print(text[0])

c) print the last character.

text = "Python Programming"
print(text[-1])

print(text[-1])

d) print "python" using slicing.

→

```
text = "python"
```

```
print(text[0:5:2])
```

~~print(text[0:6])~~

e) Count the no. of characters in the string.

→

```
text = "python programming"
```

```
print(len(text))
```

~~using for loop~~

2) list - 5 m.

Create the following list:

```
Numbers = [10, 20, 30, 40, 50]
```

write python statements to:

a) Add 60 to the list.

→

```
Numbers = [10, 20, 30, 40, 50]
```

```
Numbers.append(60)
```

```
print(Numbers)
```



b) Remove 20

→

```
Numbers = [10, 20, 30, 40, 50]
```

```
Numbers.remove(20)
```

```
print(Numbers)
```



3) change 30 to 35.

d) print the first 3 elements.

→ numbers = [10, 20, 30, 40, 50]
print(numbers[:3])

number.insert(2, 35)

e) print the length of the list.

→ print(len(numbers))

3] Tuple - SM

Create the following tuple:

fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")

write python statements to:

a) print the tuple.

→ fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")
print(fruits)

b) print "Mango".

→ print(fruits[2])

c) print the first two elements.

→ print(fruits[:2])

d) find the length of the tuple.

→ print(len(fruits))

e) check whether "Apple" exists in the tuple.

→

4) Set.

Consider:

numbers = {10, 20, 30, 20, 40, 10, 50}

Ans. the following:

a). What happens to duplicate values in a set?

→: Duplicate values in a set are automatically removed.

b). Add 60 to the set.

→: we use the add() method.

numbers.add(60)

c). Remove 30 from the set.

→: use remove() method

set.remove(30)

d). find the number of ele. in the set.

→: Use the len() fun.

5) Dictionary -

create the following dictionary:

Student = {"name": "Rahul", "age": 22, "course": "Python", "city": "Delhi"}

write python statements to:

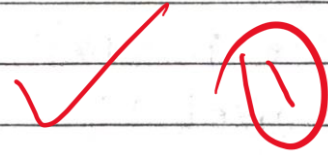
a) print the student's name.

→: student = {"name": "Rahul", "age": 22, "course": "Python", "city": "Delhi"}

Print(student["name"])

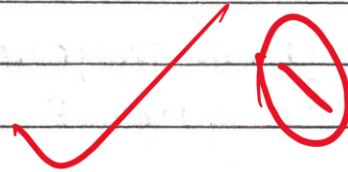
b) print the student's age.

→: `print(student["age"])`



c) Display all keys of the dictionary.

→: `print(student.keys())`

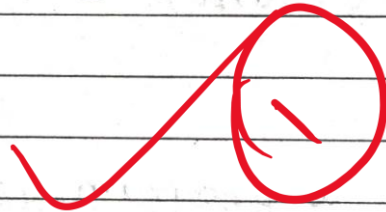


* Loops -

A] for loop -

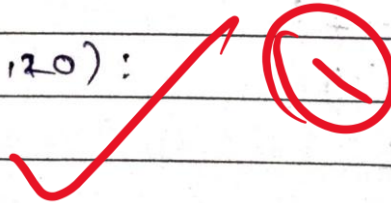
a) Write a python program using a for loop to print no's from 1 to 10.

→: `for i in range(1,11):`
`print(i)`



b) Write a python program using a for loop to print all even no's from 1 to 20.

→: `for i in range(1,20):`
`if i % 2 == 0:`
`print(i)`



B] While loop -

a) Write a python program using a while loop to print numbers from 1 to 10.

→:

→: `i = 1`
`while i <= 10:`
 `print(i)`
 `i = + 1`

b) Write a python program using a while loop to calculate the sum of no's from 1 to 10.

→: `i = 1`
`sum = 0`
`while i <= 10:`
 `sum = sum + i`
 `i = + 1`
 `print(sum)`

c) Write a program using a while loop to print no's from 10 to 1.

→: `i = 10`
`while i >= 1:`
 `print(i)`
 `i = - 1`